# 7.5 Answers: percentages

← Unit 7.5 · Practice

## Set 1 · Warm-up: what percent means

1. 40 (100% is all of it)
2. 20 (half)
3. € 6
4. 25
5. 20 (a quarter: half of 40)
6. 7 (: 10)
7. 35 (: 10)
8. 4 (: 100)
9. 50%
10. The whole pizza (1 pizza)

Talk about it: divide by 10.

## Set 2 · Building: percentages and fractions

1. 1/2
2. 1/4
3. 3/4
4. 1/10
5. 1/5 (20/100, simplified)
6. 20%
7. 30%
8. 1%
9. 40% (1/5 = 20%, so 2/5 = 40%)
10. About 33% (exactly 33 1/3%)
11. 100%

Talk about it: 100 can be shared into 5 equal parts (20 each) but not into 3 equal whole parts.

## Set 3 · Building: building from 50%, 10% and 1%

1. 12 (10% = 6, so 20% = 12)
2. 15 (10% = 5, × 3)
3. 4 (10% = 8, half of that)
4. 30 (50% = 20, 25% = 10)
5. 30 (10% = 20, 5% = 10)
6. 10 (10% = 2,5, × 4; or 40% = 2/5 and 1/5 of 25 is 5)
7. 45 (100% − 10% = 50 − 5)
8. 36 (1% = 3, × 12; or 10% = 30 and 2% = 6)
9. 27 (10% = 4,5, × 6; or 50% = 22,5 plus 10% = 4,5)
10. 5

## Set 4 · Building: discount

1. Save € 10, pay € 10
2. Save € 15, pay € 45
3. Save € 4, pay € 36
4. Save € 50, pay € 200 (10% = € 25, × 2)

5. Save € 24, pay € 56 (10% = € 8, × 3)
6. Save € 4,50, pay € 40,50
7. a. € 4,50 (€ 3 + € 1,50) b. 25% (you save € 1,50 out of € 6, which is a quarter)
8. 500 g (25% of 400 g = 100 g extra)
9. € 5

Talk about it: no. Extra means you get more for the same price; korting means you pay less for the same amount.

## Set 5 · Stretch: word problems and deals

1. 80% (20 out of 25 = 80 out of 100, multiply both by 4)
2. 75% (30/40 = 3/4)
3. 45%
4. Shop A: € 24 (Shop B is € 25,20)
5. € 9 (10% = € 1,50; 40% = € 6; € 15 − € 6)
6. 30 pupils (10% = 20, 5% = 10)
7. False. € 100 → € 50 → € 25. The second 50% is half of the *new* price, so you still pay 25% of the original.
8. 35 g (1% of 1 000 g is 10 g; 3,5 × 10 g)
9. 10

Talk about it: no. It depends on the starting prices too.

## Puzzle corner

10% of 50 = 5 and 50% of 10 = 5: the same. It always works: a% of b = b% of a, because both mean a × b : 100. So 8% of 25 = 25% of 8 = 2.